

# Score-constrained ensembles and population transport

Restricting the ordinary action of two classical phase sheets to the shared-score preparations gives the canonical Fisher field action exactly. The restriction is not preserved by free Newtonian transport. A separate population-transfer construction identifies its full transport-conjugate action; bridge phase alone omits reservoir costs. The resulting two-coordinate model satisfies continuity, but its shape family fails the full field equation, even after adding Gaussian width.

These are variational identities and a stated closure obstruction, developed by GPT-6.1 Sol and GPT-6 Astra on 2026-10-01. They do not derive a physical constraint mechanism, quantum recording, or positive action necessity. The supplied parameter is  $\kappa \geq 0$ ; the field obstruction below concerns  $\kappa > 0$ . Proof status: written, internally checked.

## 1. What the classical sheet action supplies

Work on a connected position interval or the line. Require smooth positive densities, finite displayed energies, and vanishing boundary terms for the variations used below. A classical phase sheet has position density  $r$  and momentum  $W_q$ . Its canonical one-form and free Hamiltonian are  $\int W \delta r dq$  and  $\int r W_q^2 / (2m) dq$ , respectively, with  $m > 0$ .

This one-form follows from the ordinary particle form. If material coordinates  $a$  carry fixed probability mass,  $q = X(a)$ ,  $r dq = da$ , and  $v = \delta X \circ X^{-1}$  is a virtual displacement, then  $\delta r = -\partial_q(rv)$ . Integration by parts gives

$$\int W_q(X(a)) \delta X(a) da = \int r W_q v dq = \int W \delta r dq. \quad (1)$$

Fixed total mass makes a spatially constant shift of  $W$  a gauge. This is a smooth variational identity; no infinite-dimensional well-posedness theorem is being assumed.

For two sheets of probability mass  $1/2$ , introduce the constraint

$$r_\sigma = \rho/2, \quad W_\sigma = S + \kappa \sigma \log(\rho/\rho_{\text{ref}}), \quad \sigma \in \{+1, -1\}. \quad (2)$$

Here  $\rho$  is normalized,  $S$  has action units, and the positive constant  $\rho_{\text{ref}}$  only makes the logarithm dimensionless. Changing it adds sheetwise constants and changes no momentum. Formula (2) is the field version of the supplied positive classical preparation in the recording note, §8.22.

**Proposition 1 (exact constrained action).** Pulling the two-sheet one-form and energy back to (2) gives

$$\begin{aligned} \Theta &= \sum_\sigma \int W_\sigma \delta r_\sigma dq = \int S \delta \rho dq, \\ \mathcal{H} &= \sum_\sigma \int \frac{r_\sigma W_{\sigma,q}^2}{2m} dq = \int \frac{\rho S_q^2}{2m} dq + \frac{\kappa^2}{2m} I(\rho), \\ I(\rho) &= \int \frac{\rho_q^2}{\rho} dq. \end{aligned} \quad (3)$$

The first identity cancels the two score signs; the second expands the two squares. Thus variation of the *restricted* action  $\int [\int S \rho_t dq - \mathcal{H}] dt$  yields

$$\rho_t = -\partial_q(\rho S_q/m), \quad S_t = -\frac{S_q^2}{2m} + \frac{2\kappa^2}{m} \frac{(\sqrt{\rho})_{qq}}{\sqrt{\rho}}. \quad (4)$$

No further canonical-field bracket is needed after adopting the constraint and restricted variation. The Fisher modification is established prior art [HallReginatto2002; reading: passages pp. 3–8, especially the classical action (4) and modified action/equations (17)–(18); source companion]. The calculation here identifies which correlated sheet restriction produces that action and tests its invariance.

**Why this is not an invariant reduction.** Unrestricted classical sheet continuity gives, on (2),

$$\partial_t(r_+ - r_-) = -\frac{\kappa}{m}\partial_q^2\rho. \quad (5)$$

For  $\kappa > 0$ , no strictly positive normalized density on the whole line has  $\rho_{qq} \equiv 0$ : an affine function cannot have those properties. Consequently every such constrained preparation departs from equal sheet densities immediately somewhere. Restricted variation and ordinary free sheet transport are different dynamics. Additional constraint reactions or state changes are required, as the exact Gaussian switching test in §8.23 also demonstrates. In particular (4)'s current uses the mean momentum  $S_q$ ; it does not give both physical sheet velocities  $W_{\sigma,q}/m$  without further dynamics.

On connected positive support the labeled sheet momenta determine  $S_q$  and hence  $S$  up to a common constant. At exact vacuum, separate constants on disconnected components become invisible to the ordinary physical phase-space measure. This construction cannot carry them into a vacuum limit merely by declaring them canonical data.

## 2. Population transport includes the reservoirs

Let  $h_L, h_R$  be nonnegative smooth normalized bumps with compact, ordered, disjoint supports. Between them fix a bridge  $B = [q_L, q_R]$ . Choose a smooth positive normalized density  $\rho_0$  with finite Fisher information. Put

$$\begin{aligned} h &= h_R - h_L, & \rho_w &= \rho_0 + wh, \\ G(q) &= \int_{-\infty}^q (h_L - h_R)(y)dy, & M(w) &= \int \frac{G(q)^2}{\rho_w(q)} dq, \\ R_B &= \int_B \frac{dq}{\rho_0(q)}. \end{aligned} \quad (6)$$

Here  $w$  is probability mass transferred from left to right,  $0 \leq G \leq 1$ ,  $G = 1$  on  $B$ , and  $G$  vanishes outside the reservoirs and their connector. If  $K$  is their compact enclosing interval,  $c = \min_K \rho_0 > 0$  and  $H = \|h\|_\infty > 0$ , then  $|w| < c/(2H)$  ensures  $\rho_w \geq c/2$  on  $K$ . Thus all terms in (6) are finite, with  $M(w) \leq 2|K|/c$ , and smooth in this  $w$  range.

Vanishing exterior flux and continuity uniquely require  $j(q) = \dot{w} G(q)$ , because  $\partial_q j = -\partial_t \rho_w$ . The corresponding total transport kinetic energy is  $mM(w)\dot{w}^2/2$ . Both  $M$  and  $R_B$  have length-squared units. Smooth reservoir transport contributes positively, so  $M(w) > R_B$ .

**Proposition 2 (continuity-compatible canonical family).** Define an action field, up to a common constant, by

$$S_q(q; w, P) = \frac{PG(q)}{M(w)\rho_w(q)}. \quad (7)$$

The variable  $P$  has action units; it is not a point-particle momentum. Then

$$\begin{aligned} \Theta &= \int S \delta \rho_w dq = P dw, \\ H(w, P) &= \frac{P^2}{2mM(w)} + \frac{\kappa^2}{2m} I(\rho_w), \\ \dot{w} &= \frac{P}{mM(w)}, & \dot{P} &= \frac{P^2 M'(w)}{2mM(w)^2} - \frac{\kappa^2}{2m} \frac{dI(\rho_w)}{dw}. \end{aligned} \quad (8)$$

These local reduced Hamilton equations preserve  $H$  while  $w$  remains in the positive-density range and satisfy the ambient continuity equation exactly.

*Proof.* Since  $h = -G_q$ , integration by parts yields  $\int S h dq = \int S_q G dq = P$ . The phase energy in (3) is  $P^2/(2mM)$ , proving the Hamiltonian and its equations. Their first equation gives  $\rho_w S_q/m = \dot{w}G$ , the required current.  $\square$

In this family the actual bridge action difference is

$$\Phi_B = S(q_R) - S(q_L) = \frac{R_B}{M(w)} P. \quad (9)$$

Thus  $\Phi_B$  generally differs from the population-conjugate action  $P = \int Sh_R dq - \int Sh_L dq$ . The latter is a reservoir-weighted phase-action difference. Replacing it by bridge endpoint phases omits the reservoir motion. In coordinates  $(w, \Phi_B)$  the canonical two-form has coefficient  $M(w)/R_B$ , rather than one.

For a fixed bridge density the exact dual identities are

$$E_{\text{transport}, B} = \frac{mJ^2 R_B}{2}, \quad \inf E_{\text{phase}, B} = \frac{\Phi_B^2}{2mR_B}, \quad J = \frac{\Phi_B}{mR_B}. \quad (10)$$

The phase infimum is the bridge lemma, (131), written with  $S$  instead of  $\hbar\theta$ ; it is attained in the stated weighted  $H^1$  class. The last identity makes the two energies equal. A low-density connector makes a fixed phase difference cheap and a fixed population flux expensive. It selects no absolute action scale.

The costs add over arbitrary spatial cuts, because both integrals of  $1/\rho$  and of  $G^2/\rho$  add. This is an exact spatial transport identity, not Newton's temporal cell action. In the positive-tail family of recording (133),  $R_B$  grows as  $\epsilon^{-2}$ . If reservoir costs additionally obey  $M - R_B \leq C$  uniformly, then  $|P - \Phi_B| \leq |P|C/R_B$  and the bridge becomes asymptotically dominant. That extra bound must be checked; exact vacuum is not such a check.

### 3. Continuity does not close the full field equation

Equations (8) are exact for the restricted model. They do not imply that its fixed density family is invariant under (4)'s Hamilton–Jacobi equation. For a Gaussian base  $\rho_A = N(0, A)$  and initial  $w = P = 0$ , the full equation gives

$$\partial_t S_q = \frac{\kappa^2 q}{mA^2}. \quad (11)$$

The family (7) instead permits only a multiple of  $G/\rho_A$  at that initial time. It vanishes outside a compact interval, whereas (11) does not. This is an immediate nonzero transverse term for  $\kappa > 0$ . It cannot be removed by a spatially constant action gauge.

Adding Gaussian width supplies the missing global quadratic phase at  $w = 0$ . The next calculation tests whether that repairs closure for nonzero population transfer.

**Proposition 3 (a fixed compact transfer shape cannot close).** Fix  $m, \kappa, A > 0$  and a nonzero  $h \in C_c^\infty(\mathbb{R})$  with  $\int h = 0$ . The density family  $\rho_{A,w} = \rho_A + wh$ , allowing width variation and sufficiently small positive and negative  $w$ , cannot be locally invariant under (4) while including zero-phase-gradient states for every such  $w$ , satisfying continuity with vanishing exterior flux, and having twice differentiable parameter trajectories. The same obstruction persists with Gaussian mean motion allowed. The claim concerns an open family of preparations, not all finite-dimensional models.

*Proof.* Set  $r = \rho_A$  and  $G = -\int_{-\infty}^q h$ , so  $G$  is smooth and compactly supported. The integrated density tangents for width and transfer are  $G_A = qr/(2A)$  and  $G$ . At an initially zero phase gradient, continuity first gives  $\dot{A} = \dot{w} = 0$ . Differentiating it and using (4) requires

$$\begin{aligned} \rho_w F_w &= c_A(w) G_A + c_w(w) G, \\ F_w &= \frac{2\kappa^2}{m} \partial_q \frac{(\sqrt{\rho_w})_{qq}}{\sqrt{\rho_w}}, \quad c_A = m\ddot{A}, \quad c_w = m\ddot{w}. \end{aligned} \quad (12)$$

Outside the compact supports,  $\rho_w = r$  and  $F_w = \kappa^2 q/(mA^2)$ ; hence  $c_A(w) = 2\kappa^2/(mA)$  for every small  $w$ . At  $w = 0$  this already matches the full expression, giving  $c_w(0) = 0$ . Evaluation at any point with  $G \neq 0$  makes  $c_w(w)$  smooth, since the left side is smooth while  $\rho_w > 0$ .

Put  $u = h/r$ . Differentiating the square-root ratio gives  $\delta[(\sqrt{\rho})_{qq}/\sqrt{\rho}] = (u'' - qu'/A)/2$ . Retaining the variation of the density multiplier in (12), the necessary linearized identity is

$$u''' - \frac{q}{A}u'' - \frac{u'}{A} + \frac{q}{A^2}u = cg, \quad g = G/r, \quad c = mc'_w(0)/\kappa^2. \quad (13)$$

Since  $G' = -h$ , one has  $u = qg/A - g'$ . Substitution yields

$$\begin{aligned} -g'''' + \frac{2q}{A}g''' + \left(\frac{4}{A} - \frac{q^2}{A^2}\right)g'' - \frac{4q}{A^2}g' \\ + \left(\frac{q^2}{A^3} - \frac{1}{A^2}\right)g = cg. \end{aligned} \quad (14)$$

The function  $g$  is smooth and compactly supported because  $r > 0$ . All four initial data vanish at a point outside its support. Uniqueness for this regular fourth-order linear ODE forces  $g \equiv 0$ , hence  $h = -G' \equiv 0$ , a contradiction. Allowing the Gaussian mean adds a term proportional to  $r$  in (12); tail matching forces its coefficient to zero and leaves the same contradiction.  $\square$

The zero-phase-gradient and continuity assumptions are essential to the initial tangency test. The theorem permits isolated states, shapes depending on parameters or time, and other finite-dimensional families. It specifically rules out treating a fixed compact transfer bump plus Gaussian width as an exact free Fisher evolution. Generated shape information cannot be hidden inside that projection.

#### 4. Consequence for STATE

The canonical Fisher structure follows from a specified restriction of ordinary classical sheet action; physical preservation of that restriction remains open. Population memory must carry its full transport-conjugate action, including reservoirs, before taking a weak-bridge limit. The continuity-compatible two-coordinate family fails the full field equation, and adding Gaussian width does not repair its fixed compact transfer shape. Next lemma: a paired density/phase shape mode on the expanding Gaussian, first with exact linearized field evolution and then a positive nonlinear completion. Its tail variation must be retained; linear mode closure alone is not a physical population-history model. Reject a projected residual counted as exact closure, bridge phase used as population momentum without reservoir costs, or an unexcluded zero branch presented as positive action necessity.